

WiFi Access Point Placement

Using Gradient Descent with Backtracking Line Search

Numerical Optimization - Project Report

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Abstract

This report presents a numerical optimization approach for placing a fixed number of wireless access points (APs) inside a building in order to maximize signal coverage. The problem is formulated as a constrained nonlinear optimization problem, where the decision variables are the coordinates of the APs. A projected gradient ascent method is used to solve the problem, with the gradient estimated using central finite differences and the step size selected using a backtracking line search. The building is represented as a two-dimensional grid, allowing walls and obstacles to be included through signal attenuation. An interactive implementation is also provided to visualize the floor plan, AP positions, coverage, and optimization process. The main goal of the project is to demonstrate the application and behavior of a gradient-based numerical optimization method on a practical, non-convex problem.

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1 Introduction and Background

The placement of wireless access points is an important part of designing a WiFi network. The locations of the access points affect the area that can be covered and the quality of the received signal. When a building contains multiple rooms, corridors, and obstacles, finding suitable AP locations manually becomes more difficult.

The problem can be viewed as an optimization problem in which the positions of a fixed number of APs are changed in order to maximize a coverage measure. Similar wireless network optimization problems have been studied using different objectives and optimization methods. Coverage and signal propagation are among the most common objectives, while other approaches consider throughput and load balancing [1], channel assignment [2], or localization accuracy [3].

From an optimization perspective, AP placement can be approached in several ways. Gradient-based methods use information about how the objective changes when the decision variables are modified; for example, differentiable formulations of cellular network throughput have been optimized directly through gradient descent [4]. Classical methods such as gradient descent, gradient ascent, quasi-Newton, and conjugate-gradient methods belong to this group. Other approaches are more suitable when the problem is discrete or when gradient information is not available. These include direct-search methods and metaheuristics such as genetic algorithms, simulated annealing, and tabu search [5].

For this project, a gradient-based approach was selected because it provides a direct way to study several concepts from numerical optimization. In particular, the project considers numerical gradient estimation, constrained optimization through projection, and backtracking line search for selecting the step size. Since the resulting problem is non-convex, the algorithm is expected to find a local optimum rather than necessarily the global optimum.

1.1 Related implementations

Several existing implementations were considered when designing the problem and its visualization.

- **WSN-Optimization-using-GA** uses a genetic algorithm for wireless sensor network placement and provides a useful comparison with a gradient-based approach [6].
- **plexus-network-planner** was used as inspiration for the interactive floor-plan visualization, particularly the idea of displaying AP positions and coverage directly on a building layout [7].
- **Wireless-Access-Point-Optimization** was considered as a reference for AP placement and coverage calculation [8].

These projects were used as references for the general problem and visualization. The optimization procedure used in this project was implemented separately.

2 Problem Statement

The objective of the project is to determine the placement of a fixed number of wireless access points inside a building in order to maximize signal coverage.

The decision variables are the coordinates of the APs. For n access points, the optimization variable is

$$u = (x_1, y_1, \dots, x_n, y_n),$$

where (x_i, y_i) represents the position of access point i .

The problem is constrained because an AP cannot be placed outside the building or inside a wall or obstacle. The resulting optimization landscape is non-convex and may contain multiple local optima. This means that different initial AP configurations can lead to different final solutions.

3 Optimization Model

3.1 Floor-plan representation

The building is represented using a two-dimensional grid. Each grid cell describes a part of the environment and is classified as **Outside**, **Free**, **Wall**, or **Obstacle**. Only **Free** cells are valid positions for an AP.

This representation allows the building layout to be used directly by the optimization procedure. Different floor plans can be generated with different spatial structures, including open rooms, corridors, L-shaped layouts, and multi-room office-complex layouts (kitchen, meeting room, and open workspace connected through partition doorways). This makes it possible to test the optimization method under different constraints and non-convex geometries.

Each cell also contains an attenuation coefficient $\alpha(g_x, g_y) \geq 0$, which represents the effect of walls or other obstacles on the signal.

3.2 Coverage objective

The signal contributions from multiple APs are combined to obtain the coverage of a grid point. Assuming independent coverage contributions, the probability that a point q is covered by at least one AP is

$$C(u, q) = 1 - \prod_{i=1}^n (1 - s_i(u, q)).$$

The total coverage over all free cells $\mathcal{Q}$ is then

$$g(u) = \sum_{q \in \mathcal{Q}} C(u, q).$$

To discourage APs from being placed too close to each other, a repulsion penalty is added. The minimum desired distance between two APs is defined as

$$d_{\min} = 0.8R(n).$$

The final objective function is

$$f(u) = g(u) - \sum_{i < j} 2(d_{\min} - \|u_i - u_j\|_2)^2 \mathbb{I}[0 < \|u_i - u_j\|_2 < d_{\min}].$$

The optimization problem is therefore

$$\max_{u \in \mathcal{F}} f(u),$$

where $\mathcal{F}$ represents all valid AP placements inside the building.

The objective is non-convex because of the nonlinear signal model, coverage combination, and repulsion term. In addition, the feasible region created by the floor plan is generally non-convex and

can contain disconnected areas. As a result, the optimization method is not expected to provide a guarantee of a global optimum.

4 Numerical Method

4.1 Signal model

Since AP and evaluation points are continuous but the floor plan is discretized on a grid, the set of cells crossed by a line-of-sight ray is found using Bresenham's line-rasterization algorithm [9], the standard technique for this problem in computer graphics. The total attenuation along this line is calculated as

$$A(p, q) = \sum_{(g_x, g_y) \in \text{Bresenham}(p, q)} \alpha(g_x, g_y).$$

This provides a simple way of taking walls and obstacles into account when calculating the signal received at a particular grid cell.

Each AP transmits at a fixed power, and the total power budget P_{tot} available to the network is shared equally between the n APs, so each AP transmits at P_{tot}/n . The effective range of a single AP is defined relative to a reference configuration: P_0 is the reference transmit power of one AP, and R_0 is the range that AP would achieve at that reference power. Since received power decreases with the square of distance, the range an AP can reach scales with the square root of its transmit power relative to this reference, so the effective range when n APs share the power budget is

$$R(n) = R_0 \sqrt{\frac{P_{\text{tot}}/n}{P_0}}.$$

Increasing the number of APs therefore reduces the effective range of each individual AP, reflecting the fact that the same total power is now split among more transmitters.

For an AP at position u_i and a grid point q , the distance is

$$d_i = \|q - u_i\|_2.$$

The signal contribution of AP i is modeled as

$$s_i(u, q) = \frac{1}{1 + \exp(k(d_i - 0.75R(n)))} e^{-A(u_i, q)}, \quad k = \frac{6}{R(n)}.$$

The first part of the expression models the decrease in signal with distance, while the second part reduces the signal according to the attenuation caused by the floor-plan obstacles.

4.2 Projection onto the feasible region

A normal gradient step can move an AP into a wall or outside the building. To handle this constraint, the updated position is projected onto the feasible region after every gradient step.

The projection is implemented using the floor-plan grid. Coordinates are first restricted to the floor-plan boundaries. If the resulting point is already inside a **Free** cell, it is kept unchanged. Otherwise, nearby cells are searched in expanding Chebyshev rings until a **Free** cell is found. The AP is then moved to the center of that cell.

This is an approximate projection rather than an exact Euclidean projection, but it provides a simple way to keep every accepted AP position inside the valid part of the building.

4.3 Backtracking line search

The gradient provides the direction of the update, but an appropriate step size still has to be selected. A backtracking line search is used for this purpose.

Starting with an initial step size α_0 , the step is repeatedly reduced according to

$$\alpha \leftarrow 0.7\alpha$$

until the projected candidate satisfies the sufficient-increase condition

$$u^+(\alpha) = \Pi_{\mathcal{F}}(u + \alpha \nabla f(u)),$$

$$f(u^+(\alpha)) \geq f(u) + c_1 \alpha \|\nabla f(u)\|_2^2.$$

The projection is performed before evaluating the condition. Therefore, a candidate is only accepted after it has been moved back into the feasible region.

4.4 Stopping criteria

The optimization stops when one of the following conditions is satisfied:

- the gradient norm becomes smaller than 10^{-4} ;
- the improvement in the objective becomes smaller than 10^{-6} ;
- no suitable step size can be found by the line search; or
- the maximum number of 200 iterations is reached.

The objective value and step size are recorded during the optimization so that the convergence behavior can be inspected afterwards.

Algorithm 1 Projected Gradient Ascent with Backtracking Line Search

```
1:  $u \leftarrow \Pi_{\mathcal{F}}(u_0)$ 
2:  $v \leftarrow f(u)$ 
3: for  $k = 0, \dots, K_{\max} - 1$  do
4:    $\nabla \leftarrow \text{CentralDifferenceGradient}(f, u, \varepsilon)$ 
5:   if  $\|\nabla\|_2 < \tau_g$  then
6:     stop (GradientNorm)
7:   end if
8:    $\alpha \leftarrow \alpha_0$ 
9:   found  $\leftarrow$  false
10:  for  $b = 1, \dots, B_{\max}$  do
11:     $u^+ \leftarrow \Pi_{\mathcal{F}}(u + \alpha \nabla)$ 
12:     $v^+ \leftarrow f(u^+)$ 
13:    if  $v^+ \geq v + c_1 \alpha \|\nabla\|_2^2$  then
14:      found  $\leftarrow$  true
15:      break
16:    end if
17:     $\alpha \leftarrow 0.7 \alpha$ 
18:  end for
19:  if not found then
20:    stop (LineSearchFailed)
21:  end if
22:   $u \leftarrow u^+$ 
23:   $v \leftarrow v^+$ 
24:  if  $v^+ - v < \tau_f$  then
25:    stop (ObjectiveStall)
26:  end if
27: end for
28: return  $u, v$ , iteration count, stopping reason, histories
```

4.5 Computational cost

Let $|\mathcal{Q}|$ denote the number of free cells and L the average length of a rasterized line between an AP and a grid cell. One evaluation of the coverage objective requires approximately

$$O(n|\mathcal{Q}|L)$$

operations for the signal calculations and $O(n^2)$ operations for the repulsion term.

There are $2n$ decision variables, and central finite differences require two objective evaluations for each variable. Therefore, one gradient calculation requires approximately $4n$ objective evaluations. The computational cost increases with both the number of APs and the resolution of the floor-plan grid.

5 Implementation

The project separates the numerical optimization core from the graphical interface used to visualize it. The floor plan, signal and coverage calculation, gradient estimation, projection, and optimization loop are implemented independently of any GUI code and operate purely on dense matrices (natID)

and coordinate data. The interactive visualization layer, built with the `natid` GUI framework, only wraps this core: it displays the floor plan, current AP positions, and coverage information, and drives the optimizer through the same public interface used by any other caller. This separation means the optimization procedure can be exercised, tested, or reused independently of the GUI.

The floor plan is generated as a two-dimensional grid, where each cell contains information about whether it can be occupied and, when appropriate, its attenuation value. The generator can create several structured layouts, such as open rooms, corridors, L-shaped environments, and multi-room office-complex layouts. These different layouts provide benchmark instances with different feasible regions and obstacle configurations.

The generated floor plan is then passed directly to the optimization pipeline. This allows the same optimization method to be tested under different spatial constraints without changing the numerical procedure.

For the grid representation, the occupancy and attenuation data are stored using dense matrix structures. This also makes it possible to save and load the attenuation data in MatrixMarket format when needed during development or analysis.

The interactive part of the application displays the floor plan, current AP positions, and coverage information. The optimization history can also be used to observe how the objective and step size change during the iterations. The GUI is therefore mainly used to make the numerical method easier to inspect and understand.

6 Optimization Behavior

Because the optimization problem is non-convex, the starting positions of the APs can influence the final result. Starting the algorithm from different feasible configurations can lead to different local optima.

This behavior can be observed using the interactive implementation. The AP positions can be randomized and the optimization can then be run again. Comparing different runs makes it possible to see how the initial configuration affects the final AP positions and the obtained coverage.

The visualization is particularly useful for understanding the optimization process because the changes are not limited to the final objective value. The APs can be observed as they move through the feasible region, while the objective and step-size histories provide additional numerical information about the iterations.

This also illustrates an important limitation of the chosen method. Projected gradient ascent uses local gradient information, so it does not guarantee that the global maximum will be found. For a non-convex problem such as this one, different initial configurations can result in different solutions.

7 Conclusion

This project formulates WiFi access point placement as a constrained, nonlinear optimization problem in which the coordinates of a fixed number of APs are optimized to maximize signal coverage. The building is represented as a two-dimensional grid, allowing the optimization to take walls, obstacles, and different floor-plan geometries into account.

The problem is solved using projected gradient ascent with a numerical gradient and backtracking line search. Since the feasible region is defined by the floor-plan grid, a grid-based projection is used to keep the APs inside valid locations. The backtracking line search adapts the step size

during the optimization, while the stopping criteria provide several ways of detecting convergence or termination.

The implementation also provides an interactive way to observe the behavior of the algorithm. In particular, changing the initial AP positions demonstrates the effect of non-convexity and the possibility of converging to different local optima.

Overall, the project demonstrates the use of a gradient-based numerical optimization method on a practical constrained problem. The main focus is not on finding a universally optimal WiFi placement strategy, but on showing how gradient estimation, projection, line search, and convergence criteria can be combined to solve and analyze a non-convex optimization problem.

The application is cross-platform and runs natively on macOS, Linux, and Windows. Pre-built installers for all three platforms are published as GitHub releases¹, allowing the tool to be tried directly without building it from source.

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¹https://github.com/maksimOvic/ProjNO_WAPPlacement_Maksimovic/releases